Matching Evidence to Decisions in Prolonged Crises

Can we say beforehand how far a finding travels, and for which decision?

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> **The choice.** After repeated drought and displacement, should partners keep short-term transfers, link people to a continuing safety net, add livelihood support, or build a system that can scale before the next shock?

- Fragile and conflict-affected settings: a growing share of the world's extreme poor (World Bank, 2025 projection)
- Forced displacement is long-lasting: about seven in ten refugees are in protracted situations (UNHCR, 2025)
- Humanitarian need is large and persistent: about 239 million people needing aid (OCHA, 2026)

The nexus, who decides, and how much room they have

The nexus (my wording): coordinating short-term life-saving aid with longer-term development where shocks repeat or crises last for years.

- Immediate response (where most decision studies sit) versus **programme design in protracted or repeated crises** (this agenda)
- **Planner discretion:** the choice left after rules, budget, mandate and norms. An overlooked fact on the ground, to be measured, not assumed. Advice that ignores it tends to fail (the limited uptake of cash transfers despite the evidence)

Stakeholder	Decides
Field planners	Design and adaptation
Headquarters, central research offices	Policies that constrain field choices; research agendas
Evidence funders	Which new studies to fund
Researchers	What to study
Donors, governments	Funding and rules

Evidence is worth seeking only if it could change the choice

- Three conditions: **room to choose**, **uncertainty** about the best option, and **evidence relevant enough** to move the planner's belief
- Relevance has two parts: **distance** (will it hold here?) and **actionability** (can this planner implement it?)

Where a decision sits	Planner has discretion
Relevant evidence exists and is usable	**Use it**
Evidence exists but is too distant	**Gap: transport it**
No relevant evidence	**Gap: generate or substitute**

- The research question is the **share of decisions in each row**, for each sector and decision type
- Same logic as value of information (health economics), adaptive management and decisions under deep uncertainty

Where this sits in the literature

Field	What it gives us	What it leaves open here
Generalizing from impact evaluations	Effects vary by setting and implementer; transport error can exceed local bias; hierarchical models of cross-site variation	Mostly stable settings; few comparable sites in crises
Transportability and data fusion (causal inference)	When an effect can be carried over; bounds when it cannot	Rarely tested on real programme decisions
Evidence use and uptake (mostly qualitative)	Why evidence is or is not used	Seldom measures discretion and relevance decision by decision
Evidence synthesis and gap maps (3ie)	Where studies exist	Not whether they are relevant to a given decision
Decisions under uncertainty, incl. climate and disaster response	Value of information; robust decision-making; adaptive pathways	Not applied to auditing the causal evidence a design rests on

- Nearest humanitarian work found: IDinsight (value of information for whether to commission an evaluation), ALNAP (immediate-response decisions), Elrha (barriers). None audits existing evidence against a nexus design decision.
- **Why start here:** the hard case has the least evidence and most constraint. A method that works here should carry to fields with more and closer evidence.

Three cases: evidence exists, does not decide, and travels unevenly

Kenya HSNP Phase 2

Seven expansion options differing in scale, speed and financing. The evidence supported all of them and did not speak to the choice planners faced.

A drought case; options differ in scale, not intervention type.

Two Kenyan cash studies

A randomized trial of one-time transfers (Egger et al.) and the HSNP Phase 2 local-economy evaluation both report spillovers, with different results.

Measures are defined differently; why they differ is not established. Both are on our replication list.

Bedoya et al., Afghanistan

A graduation programme tested in conflict after success in stable settings; pre-registered; large effects reported.

The aspirational case: evidence can travel. What made it possible?

No replication data found.

The mismatch is one of relevance, not price: three questions

> Evidence and research attention are largely **non-market goods**. What gets studied can be misallocated, while planners still lack evidence they can use.

- **RQ1:** how much **discretion** do planners have on a decision, and how do they **value** the evidence available (perceived relevance and usefulness)?
- **RQ2:** what do their plans **rest on** (context data, implicit knowledge, best practice, procedures), and is the barrier **searching** for evidence or **overload**?
- **RQ3:** can the barriers be reduced (lower search cost, synthesis, **least-distance evidence**), and can predictions be stated without overstating transfer and **tested prospectively**?

Goal: more successful interventions, judged by whether pre-registered predictions turn out accurate. Also separates genuine research gaps from gaps that existing evidence can fill.

RQ1: measure discretion and the value placed on evidence

- Four questions per decision: is there **discretion left** after constraints? Can **existing information** already separate the options? Does the **timeline** allow more? Who bears the **cost of being wrong**, and is it reversible?
- **First instrument: a desk audit** of a documented programme, from documents, no human subjects
  - includes whether decision-makers sought **outside support** for evidence, and under what conditions (revealed demand)
- **Later:** a documented partner funnel (approached, responded, accepted, fit), where refusals are data
- In scope: headquarters versus field constraints; where evidence generation should be allocated

> **Still open:** the taxonomy of decisions (immediate response, long-term planning, operational, design), and the headquarters/field relationship.

RQ2: what plans rest on

- Plans rest on **context data, tacit knowledge, best practice and standard procedures**
- The unit of study: the **assumptions** in causal maps, scenarios and log-frames, stated or not
- Two kinds of barrier:
  - **search:** the relevant evidence is hard to find
  - **overload:** lots of evidence, few strong signals

> **Feeds RQ3.** The causal chain, blockers and log-frame assumptions are the input to the transport model: they say which links a source study must be checked on, and the planner reviews them at each staged deliverable.

Eliciting causal maps from practitioners is the established field of **systems mapping**, usually qualitative. Linking it to DAGs for quantitative transport is the untested part.

RQ3: how far does a finding travel? What counts as comparable

- Source and target differ along **named dimensions:** spatial (including environment), temporal, institutional, market, treatment version, population, interference
- **Comparable is a profile, not one score.** On each dimension that the planner's causal map says matters:
  - **Objective:** a measured distance (e.g. the two areas' vegetation index has a similar distribution)
  - **Subjective:** the planner's similarity rating (e.g. "forest health: 4 on a 6-point scale")
- Comparable for a decision when both fall inside thresholds the planner accepts; a gap between the two is flagged
- Idea for choosing cases: an insecure region may resemble an insecure region elsewhere more than the rest of its own country

The scale is illustrative. More than one expert rating is ideal, subject to availability.

The pipeline already under construction: the 2×2

	1. Find and describe what is known	2. Test what can be used here
Evidence	**1.1 Discovery, synthesis, extraction.** Automated literature search; structured extraction of study facts and causal claims. *Built and piloted*	**2.1 Evidence gap map.** Follows the **3ie evidence gap map**, extended with context: shock, crisis phase, delivery institution, displacement, horizon. *In design*
Local, replicable	**1.2 Context data panel.** Fuses conflict, rainfall, prices, displacement, boundaries via open-source APIs and scripted or agent-assisted fetching; new country by configuration. *Built: Somalia and Sudan*	**2.2 Reproducible evaluations.** Extract and replicate eligible studies. `reprokit` *released; candidates: Egger, HSNP 2, Bangladesh insurance*

- **The loop:** the gap map shows where evidence is missing, which directs the search; extraction makes comparable records; the context panel measures closeness; replication checks which estimates reproduce; results update the map. Each pass narrows the next.
- `reprokit` : recorded hashes, rebuilds in continuous integration, so anyone can re-run a result
- Replication is fast when geodata are public (or when we replicate without them); slow only if locations must be requested privately
- A **proof of concept for a re-runnable, open-source pipeline**, not a one-off study; the dissertation validates it on one applied case

The transport model: built with the decision-maker

1. **RQ2 supplies the target map:** causal chain, blockers, log-frame thresholds
2. **Match the maps:** the source study's causal story is matched to the planner's, so both use the same variables
3. **Flag the links that differ** (a selection diagram: the causal graph with a mark on each link that works differently in the two settings)
4. **Measure and judge:** context data give the objective distance; the planner gives the similarity rating
5. **Compute a prediction** of one of four types
6. **Translate** to the planner's area, population and scale, as a target metric they can check; they correct the map and the cycle repeats

Four prediction types

Type	When
Point estimate with standard errors	Maps match, distance small
Bounds (Gechter)	Partial overlap
Ordinal category or sign	Only direction supported
"Insufficient evidence" or "distance too great"	No defensible transport: a result

> **Not a black-box predictor.** The person who will act on the answer judges what is close enough, using a map they helped draw.

Approaches to explore, and the context data behind them

Group	Approach	Where the expert enters
What can be carried over?	**Selection diagrams**	The graph comes from the planner's map
How to combine evidence	**Hierarchical Bayesian models** (partial pooling across sites)	Priors; which sites count as exchangeable
How to combine evidence	**Similarity-weighted borrowing** (reweighting; synthetic-control-style weights)	Which variables, and how close is close enough
What if it cannot?	**Partial-identification bounds**	The assumptions the bounds rest on

- **Context data:** remote sensing (rainfall, vegetation indices from Landsat, MODIS and Sentinel-2, night lights) plus conflict events, prices, displacement, boundaries
- **Limits:** satellites describe environment and activity, not treatment or welfare; resolution must suit the unit; site coordinates decide what can be linked
- **Test:** the Egger versus HSNP pair. Do the approaches agree on whether a result carried from one setting to the other?

Progress, next steps, and how it will be judged

Done so far

- Initial novelty check of the three questions against the nearest prior art (a spot check)
- Components 1.1, 1.2 and `reprokit` built
- Replication candidates identified; papers being transcribed

Next

- Desk audit of one documented programme (no human subjects)
- Replicate and compare the Kenyan cash studies
- Extend to the gap map (2.1) and the replication quadrant (2.2)
- **Outreach to J-PAL** on decisions and evidence gaps in crisis settings, and replication data access

Gates: case and partner fit → is new evidence worth it? → what can be inferred? → can it be evaluated?

Judged by: RQ1, a faculty-approved audit paper; RQ2, a map accepted as useful; RQ3, a method with empirical validation, then a partner using it

Timeline: peer talk now; faculty talk April 2027; desk audit and ethics review in year 2; thesis at end of year 3

Dates are inferred. Practitioner interviews follow ethics approval.

Why it matters beyond this case

- **Development economics:** a way to say beforehand how far an estimate travels
- **Decision science:** value of information applied to programme-design decisions, with measured discretion
- **Causal inference:** transport methods tested prospectively, with pre-registered predictions, in a hard setting
- **Climate and environmental decision-making:** a decision-relevance test for scenario and adaptation methods used under deep uncertainty
- **Research funding and synthesis:** separating real research gaps from gaps existing evidence can fill
- **Research infrastructure:** an open-source, reproducible pipeline others can re-run on a different decision
- **Humanitarian practice:** a map of what plans rest on, and where central rules block use of evidence

Intended contributions, not results. Each depends on the corresponding question working.

Appendix

A1. Selection diagrams in plain language

- An ordinary **causal graph (DAG)**: boxes for variables, arrows for cause and effect
- Add a **marker on each arrow that works differently** in the source and target settings (e.g. how payments are delivered, market access, security)
- It tells you which differences you **must measure** and whether a **formula to carry the result over exists at all**
- It cannot create data that do not exist, repair a missing outcome, or make an impossible delivery possible
- Here the graph comes from the **planner's map (RQ2)**, matched to the source study's causal story

A2. The full decision table

Relevant evidence exists?	Discretion: yes	Discretion: no (fixed by rules, budget, mandate)
Yes: relevant and actionable	**Use it**	On the shelf
Yes, but too distant	**Gap: transport the evidence (RQ3)**	Not a priority
Yes, but not actionable here	**Gap: redesign or adapt (RQ2)**	Not a priority
None	**Gap: generate or substitute**	Ignore

- Scope: decisions where the planner already has discretion and lacks relevant evidence; evidence loosening constraints is plausible but out of scope
- Other states: little demand beyond logistics; abundant evidence not used; evidence used well; planners make do; insufficient evidence either way

A3. Comparability profile (illustration)

Dimension	Objective measure	Subjective rating (1 to 6, illustrative)
Environment	Overlap of the vegetation-index distributions	Planner: forest health 4
Market	Market-access or price-pass-through indicator	Planner: 3
Institution	Delivery channel (NGO, public safety net, payment rail)	Planner: 5
Conflict exposure	Event counts and trends	Planner: 2
Population	Displacement status, composition	Planner: 4

- Thresholds per dimension are set with the planner; a gap between the two columns on a row is flagged for discussion
- Values here are placeholders, not data

A4. Validation, gates and timeline

Question	Validated when
RQ1	A working paper on the knowledge audit is approved by faculty, and at least one fit partner agrees with the funnel documented
RQ2	Desk phase: documents suffice to model constraints, scenarios and log-frames. Partner phase: the partner finds the process and products useful (needs ethics approval)
RQ3	Partial: a methodology paper with empirical validation. Full: a partner uses the results. Follow-on: predictions turn out reasonably correct

- **Gates:** case and partner fit; is new evidence worth it; what can be inferred; can it be evaluated
- **Dates (inferred):** faculty talk April 2027; ethics approval about six months; thesis end of year 3; dissertation year 5 or 6

A5. Bias controls for the feedback rounds

Someone will ask how planner feedback, with variables not fixed in advance, avoids p-hacking. The top ideas:

1. **A written, registered protocol:** the steps (data discovery, map matching, comparability profile, modelling, planner review, repeat), a stopping rule, fixed dimensions and approaches, and a log of every round and variable tried
2. **The planner rates similarity before seeing the predicted effect;** changes made after seeing the estimate are flagged
3. **Report sensitivity:** the prediction under every reasonable specification tried
4. **Lock the prediction before the outcome** (pre-registered, time-stamped)

Further controls (a planner-free baseline; more than one expert where available) are for the protocol to settle.

A6. Project maturity: built, planned, gated

Component	Status
Context panel (Somalia, Sudan)	Built and reproducing; separate config per country
`reprokit`	Released (v0.4.0, evidence kind added)
Literature discovery and extraction	Built and piloted
Evidence map (3ie-style gap map)	In design; study catalogue first
Bangladesh insurance replication	Charter and protocol filled; village coordinates are the gate for the satellite-linked step
Transferability	Planned; no repository yet
Data-fusion toolkit	Deferred

A7. Claims we keep, soften or drop

Claim	Treatment
"No one has audited evidence for programme design"	**Soften:** nearest prior art found
Egger versus HSNP as a transport failure	**Soften:** hypothesis-generating pair
A complete corpus of trials; universal impact ranges; a live simulator or dashboard	**Dropped**
Impact estimates in novel localities	**Soften** to a tiered claim
Participatory causal maps produce usable log-frames	**Proposal**
The compound-shocks panel supports transfer	**Narrowly:** context only
Evidence-use patterns across the field	**Qualitative only;** the scans were not systematic

A8. Positioning and dropped use cases

- **Prior art found:** Brown et al. 2025 and IDinsight (value of information for funding); ALNAP (immediate-response decisions); Elrha (barriers); Egger-style and Bedoya-style trials (transport and fragile-setting cases)
- **Dropped use cases:** anticipatory action and index insurance are examples of modality and timing questions only, not research strands of this agenda; operational logistics optimization is a separate adjacent agenda
- **Not a claim** about which intervention is best

A9. Open items to validate in the next few weeks

Item	Why	Next step
Scale figures (World Bank, UNHCR, OCHA)	Opening slide; found via a web-search summary	Gemini check against the primary reports; name the poverty line
Egger versus HSNP definitions	Egger's multiplier is GDP per unit transferred; HSNP's 0.38 to 0.93 may be *additional* income beyond the transfer (total about 1.38 to 1.93)	Read the HSNP local-economy methods; harmonize before any comparison
Replication feasibility	Egger's package is public but village locations are restricted; HSNP data held locally; Bedoya data not found	Confirm what each supports; ask authors only for what is needed
Bedoya et al.	Aspirational case; claims checked on the abstract and registration only	Full read; check what enabled success
Stage-2 extraction	Slide 10 says it extracts causal claims; pilot figures not re-verified	Confirm against its status doc
Literature map (slide 4)	Row descriptions are my characterization	Read one paper per field
Evidence-use scans	The "mixed, sector-dependent" picture rests on about 31 sources I have not read	Read a sample before relying on it
Nexus definition	Mine, not sourced	Read the OECD DAC recommendation
Ethics and outreach	J-PAL contact could count as human-subjects research	Ask the IRB before anything beyond public-data requests

Design questions (decision taxonomy, headquarters/field relationship, the definition of "comparable") are tracked separately.

A10. Sources consulted (1 of 2): read in full or checked against the text

- **Programme documents:** FCDO, Kenya Hunger Safety Net Programme Phase 2 — Business Case and Completion Review
- **Egger, Haushofer, Miguel, Niehaus, Walker (2022).** General equilibrium effects of cash transfers: experimental evidence from Kenya. *Econometrica*
- **Bedoya, Coville, Haushofer, Isaqzadeh, Shapiro (2019).** No household left behind: Afghanistan Targeting the Ultra Poor impact evaluation. NBER WP 25981
- **Pritchett and Sandefur (2015).** Context matters for the size of impact: experiments in other settings (microcredit)
- **Hjort et al. (2021).** How research affects policy: Brazilian municipalities
- **Crost, Felter and Johnston (2014; 2016).** Aid, insurgent response and conflict, Philippines (community-driven development; conditional cash transfers)
- **Brown, Salisbury and Ball (2025).** Value of information applied to grant allocation
- **Groh et al. (2016); Tennant et al. (2021); Altmejd et al. (2019).** Checked against the transcripts for specific claims

Full-text transcripts are held locally (git-ignored); identifiers and DOIs are in the novelty-check notes.

A11. Sources consulted (2 of 2): identifier-matched, secondhand or not yet read

- **Matched to a registry, not read:** Vivalt (2020); Allcott (site selection bias); Meager (Bayesian hierarchical analysis of microcredit); Banerjee et al. (2016, proof of concept to scalable policies); Oliver et al. (2014, barriers to evidence use); Snilstveit et al. (2016, evidence gap maps); Kwakkel, Haasnoot and Walker (2016, robust decision-making and adaptive pathways); Bareinboim and Pearl (2012, transportability)
- **Read by Gemini only:** Gechter (bounds); Montiel Olea et al. (site selection); Bongers et al. (cyclic causal models); DellaVigna, Pope and Vivalt (2019); Kastner et al. (2009) and White et al. (2020) on capture-recapture; Bold et al. (2018); IDinsight (2022); ALNAP, Campbell and Knox Clarke (2019); Elrha, Smith, Swithern and Harrison (2022)
- **Two non-systematic scans of evidence use** (about 31 sources) by Gemini: not read by me; no figures used
- **Scale figures** (World Bank 2025; UNHCR Global Trends 2025; OCHA GHO 2026): web summaries, **being verified**
- **Not read:** OECD DAC nexus recommendation; Claxton (1999); Fenwick et al. (2008)

Reference hygiene: identifiers come from registries (Crossref), not model recall.